

Finite-Sample Bounds for Multivariate Shortfall Risk Estimation: A Corrected Partial Progress Note

Prepared as a rigorous repair of an earlier draft

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Abstract

Hegde, Menon, Prashanth, and Jagannathan asked whether the finite-sample guarantees that they proved for online univariate utility-based shortfall risk can be extended to estimators of multivariate shortfall risk measures (MSRM). This note gives a corrected finite-sample analysis in two directions. First, for the compact feasible allocation set

$$\mathcal{R}_K = \{m \in K : \mathbb{E}[\ell(X - m)] \leq \tau\},$$

we prove a distribution-free empirical-process sandwich and, under an explicit margin condition, a Hausdorff bound. We also derive finite-sample bounds for the sample-average estimator of the actual MSRM capital allocation obtained by minimizing the linear capital cost over an inflated empirical feasible set. Second, for the stochastic-algorithm estimator of the KKT system used in the MSRM literature, we prove an $O(1/n)$ mean-square bound for a one-pass projected Robbins–Monro recursion under a checkable strong mean-reversion condition. A local Lyapunov/gain-matrix criterion reduces this condition to the usual Hurwitz linearization assumption plus a sufficiently small projection neighborhood.

The results below are therefore *partial progress, not a complete solution of the open problem in full generality*. They do give genuine non-asymptotic bounds for multivariate MSRM allocation estimators under strengthened local stability, boundedness, compactness, and curvature assumptions. They do not remove those assumptions, and they do not prove finite-sample bounds under only the asymptotic ODE-stability hypotheses used by Kaakaï, Matoussi, and Tamtalini.

1 Problem, scope, and main conclusion

Let X be a bounded random vector in $\mathbb{R}^d$, $d \geq 2$, interpreted as a vector of losses. Let $\ell : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex loss or aggregation function, and let $\tau \in \mathbb{R}$ be the shortfall level. Write

$$F(m) := \mathbb{E}[\ell(X - m)], \quad m \in \mathbb{R}^d. \quad (1)$$

The compactly localized acceptable allocation set is

$$\mathcal{R}_K := \{m \in K : F(m) \leq \tau\}, \quad (2)$$

where $K \subset \mathbb{R}^d$ is a nonempty compact convex eligible set. The corresponding compactly localized multivariate shortfall risk value with capital-cost vector $c \in (0, \infty)^d$ is

$$\rho_K(X) := \inf\{c^\top m : m \in K, F(m) \leq \tau\}. \quad (3)$$

The standard unweighted choice is $c = \mathbf{1}$. When $K = \mathbb{R}^d$ and $\tau = 0$, (3) is the usual “add cash then aggregate” shortfall-risk allocation problem of Armenti, Crépey, Drapeau, and Papapantoleon, except that the global problem may require additional coercivity or integrability assumptions.

The online statistical model is that $X_1, X_2, \dots$ are i.i.d. copies of X . Define the empirical shortfall surface

$$\hat{F}_n(m) := \frac{1}{n} \sum_{i=1}^n \ell(X_i - m). \quad (4)$$

What is proved. The note proves two non-asymptotic statements.

- (i) A compact SAA theorem: uniformly over $m \in K$, $\hat{F}_n(m)$ approximates $F(m)$ at rate $\sqrt{d \log n/n}$, which yields confidence sandwiches for the set (2), Hausdorff bounds under a margin condition, and an $O(\sqrt{\varepsilon_n})$ allocation error bound for the *linear-cost* MSRM allocation under strong convexity.
- (ii) A one-pass stochastic approximation theorem: the projected Robbins–Monro estimator for the KKT root of (3) satisfies $\mathbb{E}\|Z_n - z^*\|^2 = O(1/n)$ under bounded noise and a strong drift condition. This condition is locally verifiable by a gain-matrix Lyapunov argument when the Jacobian of the mean field at the KKT root is Hurwitz.

What is not proved. The preceding conclusions do not settle the broadest possible version of the open problem. In particular, the paper does not prove a global finite-sample theorem for the ungained projected stochastic algorithm under mere asymptotic ODE stability. It also does not remove compact localization or replace the curvature and strong-drift hypotheses by the weaker conditions used for asymptotic consistency and normality.

2 A finite-sample sandwich for the feasible allocation set

For $\eta \in \mathbb{R}$, define population and empirical level sets

$$\mathcal{R}_K(\eta) := \{m \in K : F(m) \leq \tau + \eta\}, \quad \hat{\mathcal{R}}_n(\eta) := \{m \in K : \hat{F}_n(m) \leq \tau + \eta\}. \quad (5)$$

Thus $\mathcal{R}_K = \mathcal{R}_K(0)$ and $\hat{\mathcal{R}}_n(0)$ is the plug-in empirical feasible set.

Assumption 2.1 (Local boundedness and Lipschitz complexity). *There are constants $R, L, B \in (0, \infty)$ such that K is contained in some Euclidean ball of radius R and, for all possible values x of X and all $m, m' \in K$,*

$$|\ell(x - m) - \ell(x - m')| \leq L\|m - m'\|_2. \quad (6)$$

Moreover the range width is at most B :

$$\sup_{x \in \text{supp}(X), m \in K} \ell(x - m) - \inf_{x \in \text{supp}(X), m \in K} \ell(x - m) \leq B. \quad (7)$$

Because X and K are bounded, Assumption 2.1 holds for every finite convex ℓ after localization to K : a finite convex function is Lipschitz on compact subsets of the interior of its domain.

For $n \geq 1$ and $\delta \in (0, 1)$ define

$$\varepsilon_n(\delta) := \frac{2B}{\sqrt{n}} + B \sqrt{\frac{d \log(1 + 2RL\sqrt{n}/B) + \log(2/\delta)}{2n}} \quad (8)$$

when $B > 0$. If $B = 0$, set $\varepsilon_n(\delta) = 0$.

Theorem 2.2 (Uniform concentration and confidence sandwich). *Under Assumption 2.1, for every $n \geq 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\sup_{m \in K} |\hat{F}_n(m) - F(m)| \leq \varepsilon_n(\delta). \quad (9)$$

On the event (9), writing $\varepsilon = \varepsilon_n(\delta)$,

$$\mathcal{R}_K(-2\varepsilon) \subseteq \hat{\mathcal{R}}_n(-\varepsilon) \subseteq \mathcal{R}_K \subseteq \hat{\mathcal{R}}_n(+\varepsilon) \subseteq \mathcal{R}_K(+2\varepsilon). \quad (10)$$

In particular,

$$\mathcal{R}_K(-\varepsilon) \subseteq \hat{\mathcal{R}}_n(0) \subseteq \mathcal{R}_K(+\varepsilon). \quad (11)$$

Proof. If $B = 0$, then every $\ell(X - m)$ is almost surely constant in the sample variable, uniformly over $m \in K$, and the claim is immediate. Assume $B > 0$. Let $r = B/(L\sqrt{n})$ and let $\mathcal{N}$ be an r -net of K . The standard Euclidean covering bound gives

$$|\mathcal{N}| \leq \left(1 + \frac{2R}{r}\right)^d = (1 + 2RL\sqrt{n}/B)^d.$$

For each $q \in \mathcal{N}$, Hoeffding's inequality and (7) imply

$$\mathbb{P}\left\{|\hat{F}_n(q) - F(q)| > t\right\} \leq 2 \exp(-2nt^2/B^2).$$

A union bound over $\mathcal{N}$ shows that, with probability at least $1 - \delta$,

$$\max_{q \in \mathcal{N}} |\hat{F}_n(q) - F(q)| \leq B \sqrt{\frac{\log(2|\mathcal{N}|/\delta)}{2n}}.$$

For any $m \in K$, choose $q \in \mathcal{N}$ with $\|m - q\|_2 \leq r$. Both F and $\hat{F}_n$ are L -Lipschitz on K , hence

$$|\hat{F}_n(m) - F(m)| \leq |\hat{F}_n(q) - F(q)| + 2Lr \leq B \sqrt{\frac{\log(2|\mathcal{N}|/\delta)}{2n}} + \frac{2B}{\sqrt{n}}.$$

Substituting the covering bound proves (9). The inclusions in (10) and (11) follow directly from the inequality $|\hat{F}_n - F| \leq \varepsilon$. $\square$

3 Metric and optimization consequences of the sandwich

Uniform objective error is not by itself a metric error bound for a level set. A margin condition is needed.

Assumption 3.1 (Two-sided sharpness). *There are constants $\gamma > 0$ and $\eta_0 > 0$ such that, for every $\eta \in [0, \eta_0]$,*

$$\mathcal{R}_K(\eta) \subseteq \mathcal{R}_K + \frac{\eta}{\gamma} B_2, \quad (12)$$

$$\mathcal{R}_K \subseteq \mathcal{R}_K(-\eta) + \frac{\eta}{\gamma} B_2. \quad (13)$$

Here B_2 is the Euclidean unit ball.

Theorem 3.2 (Hausdorff bounds). *Assume Assumptions 2.1 and 3.1. Let $\varepsilon = \varepsilon_n(\delta)$. On the event (9), the following hold for nonempty sets:*

(a) *If $\varepsilon \leq \eta_0$, then*

$$d_H(\widehat{\mathcal{R}}_n(0), \mathcal{R}_K) \leq \frac{\varepsilon}{\gamma}. \quad (14)$$

(b) *If $2\varepsilon \leq \eta_0$, then*

$$d_H(\widehat{\mathcal{R}}_n(+\varepsilon), \mathcal{R}_K) \leq \frac{2\varepsilon}{\gamma}, \quad d_H(\widehat{\mathcal{R}}_n(-\varepsilon), \mathcal{R}_K) \leq \frac{2\varepsilon}{\gamma}. \quad (15)$$

Consequently, whenever the relevant nonemptiness and sharpness conditions hold,

$$d_H(\widehat{\mathcal{R}}_n(0), \mathcal{R}_K) = O_{\mathbb{P}} \left(\frac{1}{\gamma} \sqrt{\frac{d \log n}{n}} \right).$$

Proof. The plug-in inclusions (11) give

$$\widehat{\mathcal{R}}_n(0) \subseteq \mathcal{R}_K(\varepsilon) \subseteq \mathcal{R}_K + \frac{\varepsilon}{\gamma} B_2$$

by (12), and

$$\mathcal{R}_K \subseteq \mathcal{R}_K(-\varepsilon) + \frac{\varepsilon}{\gamma} B_2 \subseteq \widehat{\mathcal{R}}_n(0) + \frac{\varepsilon}{\gamma} B_2$$

by (13). These two directed inclusions are equivalent to (14). The inflated and deflated claims use (10) and require applying Assumption 3.1 at 2ε , hence the condition $2\varepsilon \leq \eta_0$. $\square$

The next result fixes the selector in the earlier draft: the selected point is no longer the minimizer of an arbitrary strongly convex regularizer; it is the minimizer of the actual MSRM capital cost $c^\top m$.

Assumption 3.3 (Curvature and KKT multiplier for the compact MSRM allocation). *The function F is μ -strongly convex on K for some $\mu > 0$. The compact MSRM problem (3) has a unique minimizer m^* , satisfies $F(m^*) = \tau$, and admits a multiplier $\theta^* > 0$ such that*

$$m^* \in \arg \min_{m \in K} \left\{ c^\top m + \theta^*(F(m) - \tau) \right\}. \quad (16)$$

Given the confidence radius $\varepsilon_n(\delta)$, define the conservative empirical MSRM allocation

$$\widehat{m}_n \in \arg \min \left\{ c^\top m : m \in K, \widehat{F}_n(m) \leq \tau + \varepsilon_n(\delta) \right\}. \quad (17)$$

The inflation makes the true optimizer feasible on the same high-probability event used in Theorem 2.2.

Theorem 3.4 (Finite-sample bound for the compact MSRM allocation). *Assume Assumptions 2.1 and 3.3. Let $\varepsilon = \varepsilon_n(\delta)$. With probability at least $1 - \delta$, the estimator (17) is well-defined and satisfies*

$$F(\widehat{m}_n) - \tau \leq 2\varepsilon, \quad (18)$$

$$0 \leq \rho_K(X) - c^\top \widehat{m}_n \leq 2\theta^*\varepsilon, \quad (19)$$

$$\|\widehat{m}_n - m^*\|_2 \leq 2\sqrt{\varepsilon/\mu}. \quad (20)$$

In particular,

$$\|\widehat{m}_n - m^*\|_2 = O_{\mathbb{P}} \left(\left(\frac{d \log n}{n} \right)^{1/4} \right), \quad |c^\top \widehat{m}_n - \rho_K(X)| = O_{\mathbb{P}} \left(\sqrt{\frac{d \log n}{n}} \right). \quad (21)$$

Proof. Work on the event (9). Since $F(m^\star) = \tau$, we have $\hat{F}_n(m^\star) \leq \tau + \varepsilon$, so $m^\star$ is feasible for (17). Therefore $c^\top \hat{m}_n \leq c^\top m^\star = \rho_K(X)$. Also,

$$F(\hat{m}_n) \leq \hat{F}_n(\hat{m}_n) + \varepsilon \leq \tau + 2\varepsilon,$$

which proves (18).

Set

$$L_\star(m) := c^\top m + \theta^\star(F(m) - \tau).$$

By Assumption 3.3, $L_\star$ is $\theta^\star\mu$ -strongly convex on K and is minimized over K at $m^\star$. Hence

$$L_\star(m) - L_\star(m^\star) \geq \frac{\theta^\star\mu}{2} \|m - m^\star\|_2^2, \quad m \in K. \quad (22)$$

Applying (22) to $m = \hat{m}_n$ gives

$$0 \leq c^\top(\hat{m}_n - m^\star) + \theta^\star(F(\hat{m}_n) - \tau) \leq 0 + 2\theta^\star\varepsilon.$$

This implies (20). It also implies

$$c^\top m^\star - c^\top \hat{m}_n \leq \theta^\star(F(\hat{m}_n) - \tau) \leq 2\theta^\star\varepsilon,$$

which proves (19). $\square$

Remark 3.5 (Why the $n^{-1/4}$ location rate appears). *The empirical shortfall constraint is estimated at objective-level accuracy $\varepsilon_n = O_{\mathbb{P}}(\sqrt{d \log n/n})$. A strongly convex Lagrangian converts objective error into squared distance, giving a distance rate of order $\sqrt{\varepsilon_n}$. A root-finding estimator with strong mean reversion, analyzed next, can achieve an $n^{-1/2}$ root-mean-square allocation rate.*

4 A one-pass finite-sample theorem for the KKT stochastic algorithm

The preceding estimator is a sample-average estimator. The stochastic-algorithm estimator used by Kaakaï, Matoussi, and Tamtalini is instead based on a KKT root. We first state a general projected stochastic-approximation theorem, then specialize it to MSRM.

Let $p \geq 1$, let $\mathcal{K} \subset \mathbb{R}^p$ be nonempty, compact, and convex, and let $z^\star \in \mathcal{K}$. Let $H : \mathcal{X} \times \mathcal{K} \rightarrow \mathbb{R}^p$ be a measurable oracle and define

$$h(z) := \mathbb{E}[H(X, z)]. \quad (23)$$

Let $G \in \mathbb{R}^{p \times p}$ be a deterministic gain matrix. The gained projected recursion is

$$Z_{k+1} = \Pi_{\mathcal{K}}(Z_k + a_{k+1}GH(X_{k+1}, Z_k)), \quad k \geq 0, \quad (24)$$

where $\Pi_{\mathcal{K}}$ is Euclidean projection and

$$a_{k+1} = \frac{\alpha}{k + \beta + 1}, \quad \alpha > \frac{1}{2\kappa}, \quad \beta \geq 2\kappa\alpha. \quad (25)$$

Assumption 4.1 (Bounded gained oracle and strong drift). *The root $z^\star$ satisfies $h(z^\star) = 0$. There are constants $M < \infty$ and $\kappa > 0$ such that, almost surely and for every $z \in \mathcal{K}$,*

$$\|GH(X, z)\|_2 \leq M, \quad (26)$$

$$(z - z^\star)^\top Gh(z) \leq -\kappa\|z - z^\star\|_2^2. \quad (27)$$

Theorem 4.2 (Finite-sample projected Robbins–Monro bound). *Under Assumption 4.1, the recursion (24) satisfies, for all $n \geq 0$,*

$$\mathbb{E}\|Z_n - z^*\|_2^2 \leq \frac{Q}{n + \beta}, \quad Q := \max \left\{ \beta \|Z_0 - z^*\|_2^2, \frac{\alpha^2 M^2}{2\kappa\alpha - 1} \right\}. \quad (28)$$

Consequently, for every $\delta \in (0, 1)$,

$$\mathbb{P} \left\{ \|Z_n - z^*\|_2 \leq \sqrt{\frac{Q}{(n + \beta)\delta}} \right\} \geq 1 - \delta. \quad (29)$$

Proof. Let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$ and $e_k = Z_k - z^*$. Since Euclidean projection onto a closed convex set is nonexpansive and $z^* \in \mathcal{K}$,

$$\|e_{k+1}\|_2^2 \leq \|e_k + a_{k+1}GH(X_{k+1}, Z_k)\|_2^2.$$

Taking conditional expectation and using Assumption 4.1,

$$\begin{aligned} \mathbb{E}[\|e_{k+1}\|_2^2 \mid \mathcal{F}_k] &\leq \|e_k\|_2^2 + 2a_{k+1}e_k^\top Gh(Z_k) + a_{k+1}^2 M^2 \\ &\leq (1 - 2\kappa a_{k+1})\|e_k\|_2^2 + a_{k+1}^2 M^2. \end{aligned}$$

Let $u_k = \mathbb{E}\|e_k\|_2^2$, $A = 2\kappa\alpha > 1$, and $C = \alpha^2 M^2$. Then

$$u_{k+1} \leq \left(1 - \frac{A}{k + \beta + 1}\right) u_k + \frac{C}{(k + \beta + 1)^2}. \quad (30)$$

We prove by induction that $u_k \leq Q/(k + \beta)$. The claim is true at $k = 0$ by the definition of Q . Assume it at time k and put $N = k + \beta + 1$. Since $\beta \geq A$, the coefficient in (30) is nonnegative. Thus

$$u_{k+1} \leq \left(1 - \frac{A}{N}\right) \frac{Q}{N-1} + \frac{C}{N^2} = \frac{Q(N-A)}{N(N-1)} + \frac{C}{N^2}.$$

It is enough to show that the last display is at most Q/N , i.e.

$$C \leq Q \frac{N(A-1)}{N-1}.$$

This follows from $Q \geq C/(A-1)$ and $N/(N-1) \geq 1$. Markov's inequality gives (29). $\square$

4.1 Local verification by a gain-matrix Lyapunov criterion

The strong drift condition (27) is stronger than asymptotic stability, but it is locally checkable. The following proposition is the finite-sample analogue of the standard linearization step in asymptotic stochastic approximation.

Proposition 4.3 (Hurwitz linearization implies local strong drift after gaining). *Let $h : U \rightarrow \mathbb{R}^p$ be continuously differentiable on a convex neighborhood U of z^* , with $h(z^*) = 0$. Let $J := Dh(z^*)$ be Hurwitz. Let $P = P^\top \succ 0$ be the unique solution of the Lyapunov equation*

$$PJ + J^\top P = -I_p. \quad (31)$$

Let $\mathcal{K} \subset U$ be compact, convex, and contain z^ . Define*

$$\Delta_{\mathcal{K}} := \sup_{z \in \mathcal{K}} \|\text{Sym}(PDh(z)) - \text{Sym}(PJ)\|_{\text{op}}, \quad \text{Sym}(A) := \frac{A + A^\top}{2}. \quad (32)$$

If $\Delta_{\mathcal{K}} < 1/2$, then, for all $z \in \mathcal{K}$,

$$(z - z^*)^\top Ph(z) \leq -\left(\frac{1}{2} - \Delta_{\mathcal{K}}\right) \|z - z^*\|_2^2. \quad (33)$$

Thus Assumption 4.1 holds for the gain $G = P$ with $\kappa = 1/2 - \Delta_{\mathcal{K}}$, provided the gained oracle is bounded on $\mathcal{K}$.

Proof. The Lyapunov equation gives $\text{Sym}(PJ) = -I_p/2$. For $e = z - z^*$,

$$Ph(z) = \int_0^1 PDh(z^* + te)e dt.$$

Therefore

$$\begin{aligned} e^\top Ph(z) &= \int_0^1 e^\top \text{Sym}(PDh(z^* + te))e dt \\ &\leq \int_0^1 \left(-\frac{1}{2} + \Delta_{\mathcal{K}}\right) \|e\|_2^2 dt, \end{aligned}$$

which is (33). □

5 Application to the MSRM KKT estimator

Assume now that ℓ is continuously differentiable and that the compact MSRM problem (3) has a KKT pair $z^* = (m^*, \theta^*) \in \mathbb{R}^d \times (0, \infty)$. The Lagrangian is

$$\mathcal{L}(m, \theta) = c^\top m + \theta(F(m) - \tau), \quad \theta \geq 0. \quad (34)$$

Since $\nabla F(m) = -\mathbb{E}[\nabla \ell(X - m)]$, the KKT equations are

$$\theta^* \mathbb{E}[\nabla \ell(X - m^*)] = c, \quad F(m^*) = \tau. \quad (35)$$

Define $p = d + 1$ and, for $z = (m, \theta)$,

$$H(x, z) := \begin{pmatrix} \theta \nabla \ell(x - m) - c \\ \ell(x - m) - \tau \end{pmatrix}, \quad (36)$$

$$h(z) := \mathbb{E}[H(X, z)] = \begin{pmatrix} \theta \mathbb{E}[\nabla \ell(X - m)] - c \\ F(m) - \tau \end{pmatrix}. \quad (37)$$

Then (35) is exactly $h(z^*) = 0$. The one-pass gained projected KKT estimator is

$$Z_{k+1} = \Pi_{\mathcal{K}}(Z_k + a_{k+1}GH(X_{k+1}, Z_k)), \quad Z_k = (M_k, \Theta_k), \quad (38)$$

where $\mathcal{K} \subset \mathbb{R}^d \times [0, \infty)$ is a compact convex projection set containing z^* .

Corollary 5.1 (Finite-sample MSRM KKT bound under strong drift). *Suppose (36) is bounded after gaining on $\mathcal{K}$, and suppose there is a gain matrix G and a constant $\kappa > 0$ such that*

$$(z - z^*)^\top Gh(z) \leq -\kappa \|z - z^*\|_2^2, \quad z \in \mathcal{K}. \quad (39)$$

Then (38), with the stepsizes (25), satisfies

$$\mathbb{E}\|Z_n - z^\star\|_2^2 \leq \frac{Q}{n + \beta} \quad (40)$$

with Q as in Theorem 4.2. In particular,

$$\mathbb{E}\|M_n - m^\star\|_2^2 \leq \frac{Q}{n + \beta}, \quad (41)$$

$$\mathbb{E}|c^\top M_n - \rho_K(X)|^2 \leq \frac{\|c\|_2^2 Q}{n + \beta}. \quad (42)$$

The corresponding high-probability bounds follow from (29).

Proof. Apply Theorem 4.2 with the oracle (36). The allocation bound follows by projecting $Z_n - z^\star$ onto its first d coordinates. The value bound follows from Cauchy's inequality,

$$|c^\top M_n - c^\top m^\star|^2 \leq \|c\|_2^2 \|M_n - m^\star\|_2^2.$$

□

When ℓ is twice continuously differentiable, the Jacobian of (37) at $z^\star = (m^\star, \theta^\star)$ is

$$J = \begin{pmatrix} -\theta^\star A & b \\ -b^\top & 0 \end{pmatrix}, \quad A := \mathbb{E}[\nabla^2 \ell(X - m^\star)], \quad b := \mathbb{E}[\nabla \ell(X - m^\star)]. \quad (43)$$

The next statement is the promised local version of Corollary 5.1.

Corollary 5.2 (Local finite-sample bound under Hurwitz linearization). *Assume ℓ is C^2 on a neighborhood of $\{x - m : x \in \text{supp}(X), (m, \theta) \in \mathcal{K}\}$, that h in (37) is continuously differentiable on a convex neighborhood of $z^\star$, and that the Jacobian J in (43) is Hurwitz. Let $P \succ 0$ solve (31). If $\mathcal{K}$ is chosen so that $\Delta_{\mathcal{K}} < 1/2$ in (32), then the gained MSRM recursion (38) with $G = P$ satisfies the finite-sample bounds (40)–(42) with*

$$\kappa = \frac{1}{2} - \Delta_{\mathcal{K}}, \quad M = \sup_{x \in \text{supp}(X), z \in \mathcal{K}} \|PH(x, z)\|_2.$$

Proof. This is Proposition 4.3 applied to the MSRM mean field (37), followed by Corollary 5.1. Boundedness of M follows from compactness of $\mathcal{K}$, boundedness of X , and continuity of ℓ and $\nabla \ell$ on the relevant compact set. □

Remark 5.3 (Why a gain or Lyapunov argument is not cosmetic). *For the raw KKT field (37), Euclidean strong monotonicity usually fails even when the root is locally asymptotically stable. Indeed, the symmetric part of the Jacobian (43) has a zero multiplier-multiplier block. The gain matrix in Proposition 4.3 converts Hurwitz stability into a local Euclidean drift inequality for the gained mean field. This is precisely the step that is missing from a naive vector extension of the one-dimensional shortfall-risk Robbins–Monro proof.*

6 Comparison with the original draft and status of the open problem

The earlier draft contained correct ingredients, but overstated their scope. The compact concentration theorem is valid after minor correction. In particular, the Hausdorff bounds for inflated and deflated sets require $2\varepsilon_n(\delta) \leq \eta_0$, because the proof invokes the margin condition at level $2\varepsilon_n(\delta)$.

The more substantive issue was conceptual. Estimating a compact feasible set $\{F \leq \tau\}$, or choosing an arbitrary strongly convex selector from that set, is not the same as estimating the MSRM allocation, which minimizes the linear capital cost $c^\top m$ subject to the shortfall constraint. Theorem 3.4 repairs this part for a localized SAA estimator by using the actual linear objective and a KKT multiplier.

For streaming estimation, a scalar ray recursion is only a univariate root-finding problem embedded in $\mathbb{R}^d$. It does not analyze the multivariate KKT estimator. Corollaries 5.1 and 5.2 instead apply directly to the MSRM KKT root, but only under a gain/local strong-drift assumption. This is meaningful partial progress toward the question posed by Hegde, Menon, Prashanth, and Jagannathan, because it gives explicit non-asymptotic $O(1/n)$ mean-square bounds for a one-pass multivariate MSRM allocation estimator. It is not a complete answer because the strongest asymptotic results in the existing MSRM stochastic-algorithm literature assume qualitative stability/Hurwitz conditions and prove consistency/CLTs, whereas the finite-sample result here requires a quantitative local drift region and bounded gained noise.

Acknowledgment of limitations

The constants in the stochastic-approximation bound depend on a projection set containing the unknown root and, in the local theorem, on the gain matrix obtained from the unknown Jacobian. This is common in non-asymptotic stochastic-approximation theory but limits direct implementability. Adaptive gain estimation, Polyak–Ruppert averaging, and high-probability martingale bounds sharper than Markov’s inequality are natural next steps. Removing compact localization, treating heavy-tailed losses, and proving finite-sample bounds under the exact assumptions of Kaakāi–Matoussi–Tamtalini remain open from the present note’s perspective.

References

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